NOAA Technical Memorandum NMFS-XXX-##

# Caribbean Ecosystem Status Report

## Gulf of Mexico Integrated Ecosystem Assessment Program

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U.S. DEPARTMENT OF COMMERCE

National Oceanic and Atmospheric Administration
National Marine Fisheries Service
Northwest Fisheries Science Center

# Caribbean Ecosystem Status Report

Gulf of Mexico Integrated Ecosystem Assessment Program[1]

1. NOAA Fisheres, Southeast Fisheries Science Center

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# Executive Summary

TEST making a change

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# 1 Introduction

## 1.1 About this report

This report is for the Caribbean Fishery Management Council (CFMC). The purpose of this report is to synthesize ecosystem and socioeconomic information to help the CFMC better meet fishery management objectives. This report relied on both previously identified proposed indicators as well as expert vetting to select a suite of indicators that best address the fishery management plan (FMP) objectives for the U.S. Caribbean. Information in this report is organized into two sections: tracking performance toward fishery management objectives and potential risks to meeting fishery management objectives. The style of this report is based on the 2023 State of the Ecosystem Reports for the Northeast U.S. Shelf.

## 1.2 Report structure

The CFMC's Science and Statistical Committee, as well as the region's Ecosystem-Based Fishery Management Technical Advisory Panel (EBFM TAP), recently completed a series of conceptual models linking key components of the ecosystem and human activities related to fishing. This report used these conceptual models as a starting list of proposed indicators and matched the indicators to answer FMP objectives when possible. For those objectives that did not have an immediate conceptual model-identified indicator, this report used a decision matrix process for expert vetting (Figure 1.1).

This decision matrix was composed of a list of proposed indicators compiled from the conceptual models as well as proposed indicators provided via expert input. These potential indicators were vetted and edited by expert small working groups, who then scored a decision matrix (Figure 1.1) of potential indicators against the following decision criteria: long term data availability, measurability, sensitivity to environmental changes, specificity, spatial and temporal scalability, relevance to specific FMP objectives, and responsiveness to management actions.

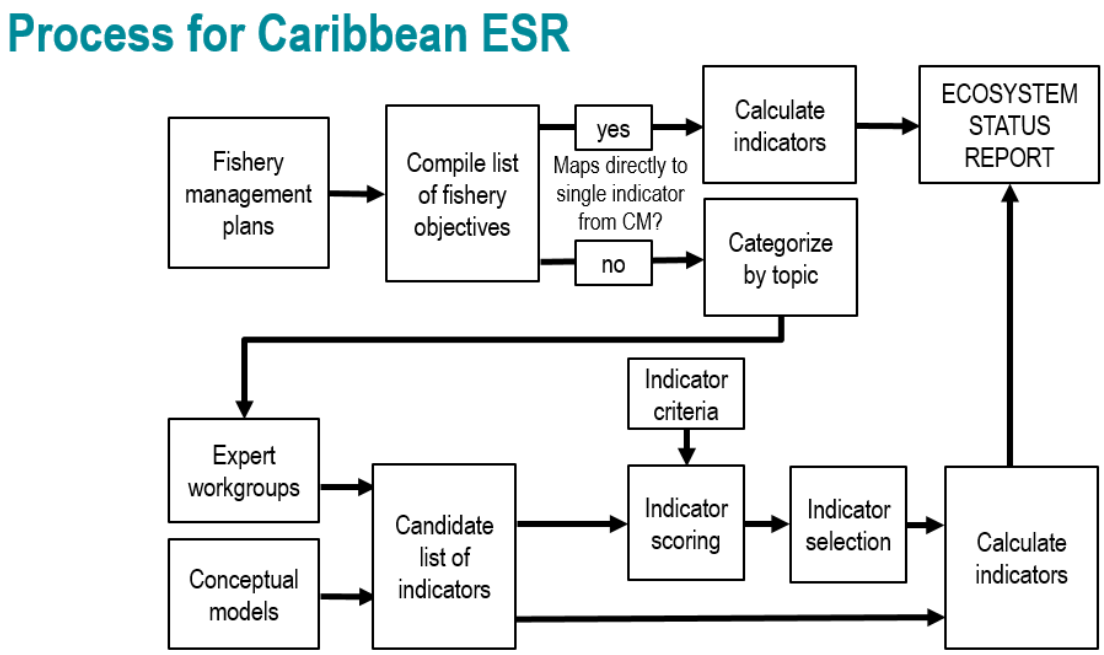


Figure 1.1: Process for selecting indicators for the U.S. Caribbean Ecosystem Status Report.

## 1.3 Notes on interpreting time series figures

Time series data are plotted in a standardized format for ease of interpretation (e.g., Figure 1.2). The x-axis represents the temporal dimension, which may be monthly, yearly, or irregular time steps, and the y-axis represents the indicator value in units specified in the axis label. The dashed horizontal line represents the mean indicator value across the entire time series, and the solid horizontal lines denote the mean plus or minus one standard deviation. Red shaded areas and green shaded areas show years for which the indicator value is below or above one standard deviation from the mean, respectively. The blue vertical shaded box highlights the last five years of indicator values, over which additional metrics are calculated. Black circles to the right of each figure indicate whether the indicator values over the last five years are greater (plus sign), less than (minus sign), or within (solid circle) one standard deviation from the mean of the overall time series. Arrows to the right of each figure indicate whether the least squares linear fit through the last five years of data produces a positive or negative slope that is greater than one standard deviation (upward or downward arrows respectively), or less than one standard deviation (left-right arrow).

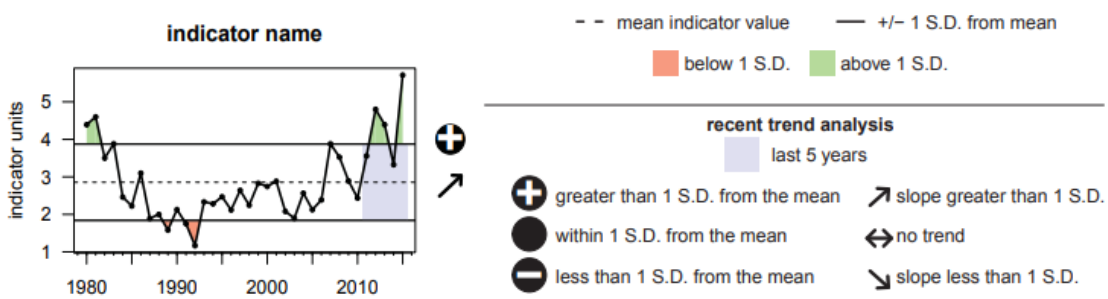


Figure 1.2: Example time series plot, showing an indicator plotted with its mean and standard deviation, and trend analysis for the most recent five years of data. See text for more detailed description of specific calculations.

# 2 Risks to meeting fishery management objectives

In this section, we examine indicators...

### 2.0.1 Coral bleaching stress

Accumulated heat stress, which can lead to coral bleaching and death, is measured by summing degree heating weeks for the most recent 12-week period from sea surface temperature data (cite). Bleaching stress was generally below average until the mid-2000s, until a sudden bleaching event in 2005 which is now the second most severe event in history. In 2024, and unprecedented bleaching event occurred across the U.S. Caribbean and beyond (Figure 2.1).

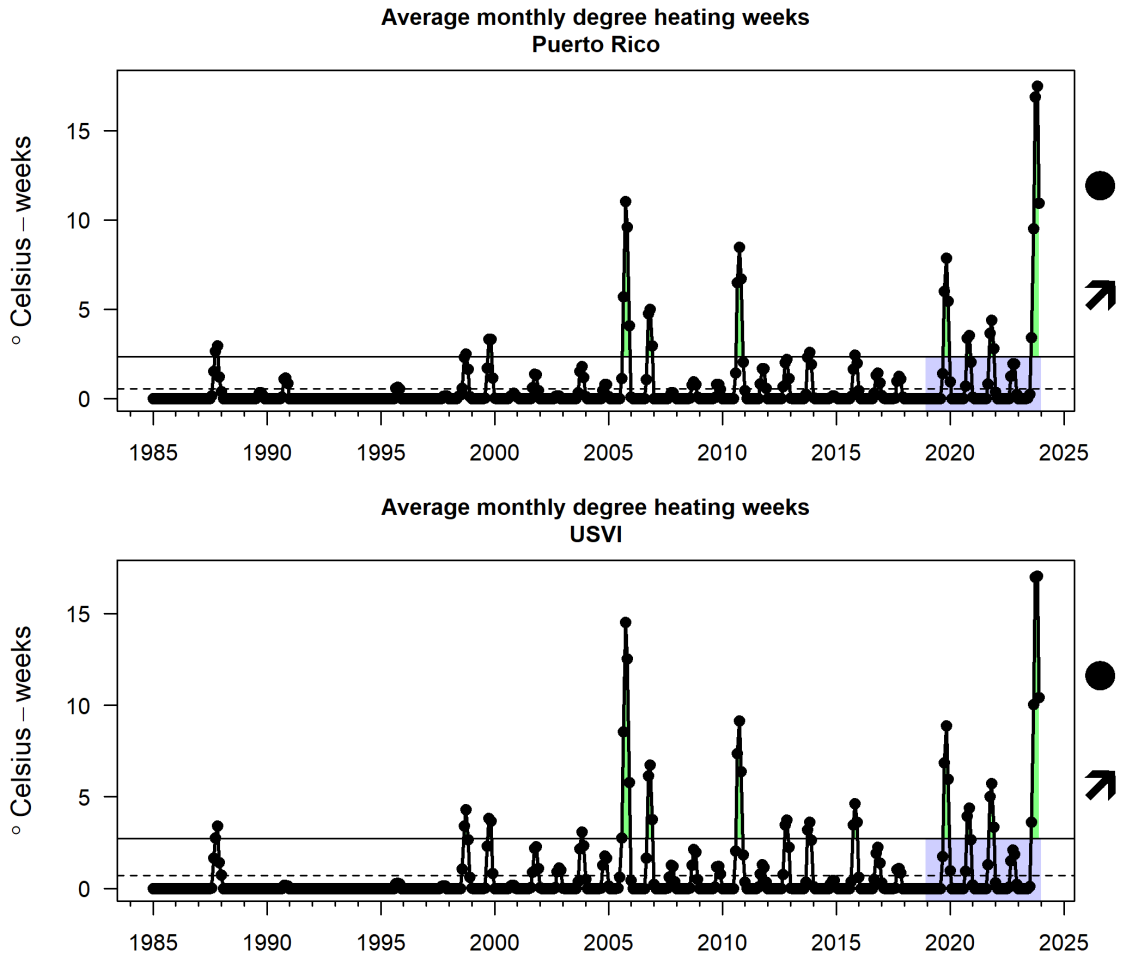


Figure 2.1: Number of degree heating weeks in Puerto Rico and the USVI as a measure of thermal stress to corals.

# 3 Customization

## 3.1 Quarto documentation

Quarto allow many bells and whistles to make nice output. Read the documentation here Quarto documentation.

## 3.2 Examples

Looking at other people's Quarto code is a great way to figure out how to do stuff. Most will have a link to a GitHub repo where you can see the raw code. Look for a link to edit page or see source code. This will usually be on the right. Or look for the GitHub icon somewhere.

- Quarto gallery
- nmfs-openscapes
- Faye lab manual
- quarto-titlepages Note the link to edit is broken. Go to repo and look in `documentation` directory.

# 4 Rendering

The repo includes a GitHub Action that will render (build) the website automatically when you make changes to the files. It will be pushed to the `gh-pages` branch.

But when you are developing your content, you will want to render it locally.

## 4.1 Step 1. Make sure you have a recent RStudio

Have you updated RStudio since about August 2022? No? Then update to a newer version of RStudio. In general, you want to keep RStudio updated and it is required to have a recent version to use Quarto.

## 4.2 Step 2. Clone and create RStudio project

First, clone the repo onto your local computer. How? You can click File > New Project and then select "Version Control". Paste in the url of the repository. That will clone the repo on to your local computer. When you make changes, you will need to push those up.

## 4.3 Step 3. Render within RStudio

RStudio will recognize that this is a Quarto project by the presence of the `_quarto.yml` file and will see the "Build" tab. Click the "Render website" button to render to the `_site` folder.

**Previewing:** You can either click `index.html` in the `_site` folder and specify "preview in browser" or set up RStudio to preview to the viewer panel. To do the latter, go to Tools > Global Options > R Markdown. Then select "Show output preview in: Viewer panel".

# 5 Figures and Tables

Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see http://rmarkdown.rstudio.com.

## 5.1 Code

You can embed an R code chunk like this:

```r
summary(cars)
```

```
     speed           dist       
 Min.   : 4.0   Min.   :  2.00  
 1st Qu.:12.0   1st Qu.: 26.00  
 Median :15.0   Median : 36.00  
 Mean   :15.4   Mean   : 42.98  
 3rd Qu.:19.0   3rd Qu.: 56.00  
 Max.   :25.0   Max.   :120.00  
```

## 5.2 Including Plots

You can also embed plots and reference them, like so Figure 5.1.

Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.

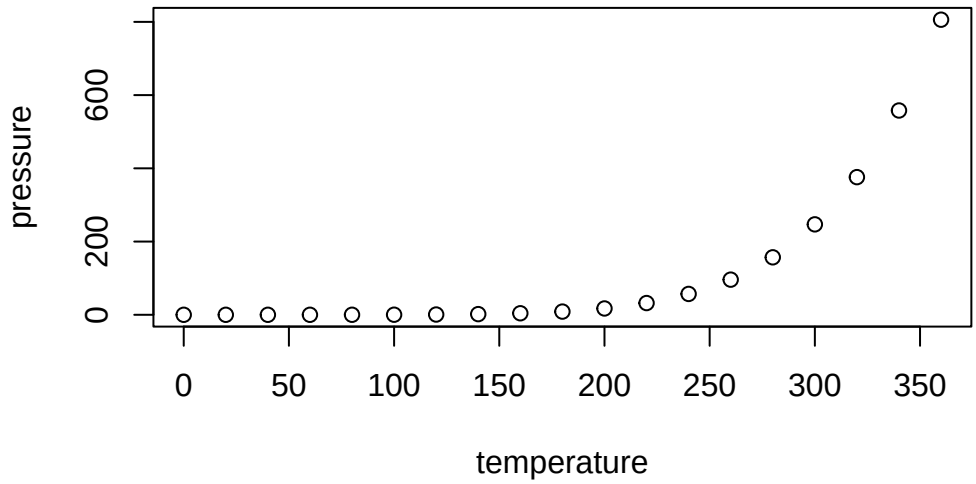


Figure 5.1: Plot of pressure

## 5.3 Including Tables

You can also embed tables and reference them with Table 5.1.

```
library(knitr)
kable(head(iris))
```

Table 5.1: Iris Data

| Sepal.Length | Sepal.Width | Petal.Length | Petal.Width | Species |
|---|---|---|---|---|
| 5.1 | 3.5 | 1.4 | 0.2 | setosa |
| 4.9 | 3.0 | 1.4 | 0.2 | setosa |
| 4.7 | 3.2 | 1.3 | 0.2 | setosa |
| 4.6 | 3.1 | 1.5 | 0.2 | setosa |
| 5.0 | 3.6 | 1.4 | 0.2 | setosa |
| 5.4 | 3.9 | 1.7 | 0.4 | setosa |

# 6 Rendering with Code

You can have code (R, Python or Julia) in your qmd file. You will need to have these installed on your local computer, but presumably you do already if you are adding code to your qmd files.

```r
x <- c(5, 15, 25, 35, 45, 55)
y <- c(5, 20, 14, 32, 22, 38)
lm(x ~ y)
```

```
Call:
lm(formula = x ~ y)

Coefficients:
(Intercept)            y
      1.056        1.326
```

## 6.1 Modify the GitHub Action

You will need to change the GitHub Action in `.github/workflows` to install these and any needed packages in order for GitHub to be able to render your webpage. The GitHub Action install R since I used that in `code.qmd`. If you use Python or Julia instead, then you will need to update the GitHub Action to install those.

If getting the GitHub Action to work is too much hassle (and that definitely happens), you can alway render locally and publish to the `gh-pages` branch. If you do this, make sure to delete or rename the GitHub Action to something like

```
render-and-publish.old_yml
```

so GitHub does not keep trying to run it. Nothing bad will happen if you don't do this, but if you are not using the action (because it keeps failing), then you don't need GitHub to run it.

## 6.2 Render locally and publish to gh-pages branch

To render locally and push up to the `gh-pages` branch, open a terminal window and then `cd` to the directory with the Quarto project. Type this in the terminal:

```
quarto render gh-pages
```

# 7 References

Quarto has powerful references functionality. You can easily insert citations from Zotero libraries that you maintain in the cloud (on Zotero). This allows the whole team to update the library and you can sync up to that library. Read about this on the Quarto documentation on citations. Google youtube videos on this also to see it in action.

Add a `.bib` file in to your project or add a linked Zotero library via RStudio in Visual mode with Tools > Project Options... > R Markdown > select custom libraries from the Zotero dropdown.

The you can type `@` and you will see a dropdown of the references in your libraries. You can then select the ones to add. If you don't see the one you need, you can paste in the DOI and it will be added to your references file (with all the info). The references will be added to your references section of your book automatically.

See the `references.qmd` file for how to include the references.

- `@ansley1981` will produce Ansley and Davis (1981)
- `[@ansley1981]` will produce (Ansley and Davis 1981).

# References

Ansley, H. L. H., and C. D. Davis. 1981. "Migration and Standing Stock of Fishes Associated with Artificial and Natural Reefs on Georgia's Outer Continental Shelf." Brunswick, Georgia, USA.